

INVESTIGATION REPORT

Issue 1005: Outer loop ending inside the inner loop

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1005
Disposition	Waiver, with a stated software rule
Record date	2026-09-11
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1005
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	COREV_PULP=1, FPU=0 (RTL master 6033d2b)
Disposition	Waiver, with a stated software rule
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	9 cited, 7 present, 2 not retained

Disposition meaning. The code is unreachable only while software obeys an explicitly stated rule. The rule is part of the disposition and is recorded in the Action section.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

cv32e40p_controller.sv:640 is 'if (hwlp_end1_eq_pc && hwlp_counter1_gt_1)' inside 'if (is_hwlp_body)' in the DECODE default arm. The uncovered condition is hwlp_end1_eq_pc==1 with hwlp_counter1_gt_1==0 while is_hwlp_body==1. Since counter1 <= 1 kills HWLoop1's own term, is_hwlp_body must come from HWLoop0 (start0 <= pc, end0 >= pc+4, counter0 > 1), so HWLoop1's end register must equal pc+4 with pc inside HWLoop0's body: the outer loop's end sits inside the inner loop, the same nesting violation as #1009 (mirror of line 630, which is covered because loop0 nested in loop1 is legal). Hardware does not check this: the cover engine reaches the exact condition in 7 cycles with an illegal register set (and address wrap at the top of memory). Under the PULP nesting rule (end1 >= end0+8, start0 >= start1) it is impossible: z3 unsat with the rule (also unsat for disjoint sequential loops), sat without it (model: end1 = pc+4 inside loop0, counter1=0). Formal proof under the invariant: a first attempt with only the #1009 register-content rule was defeated by a 32-bit wrap witness (start0 > end0 at 0xfffffff, kept as evidence); adding well-formed loops (start < end when active) and no address wrap, k-induction (k=12) proves the condition unreachable (966 s). abc pdr did not converge in 2.4 h. Caveat as in #1009: a stale HWLoop1 end register (counter1 = 0 after a finished loop) lying inside a later HWLoop0 body also reaches the arm with sequential, non-nested loops, so the waiver invariant must be read on register contents. Disposition matches the 2024 waiver but replaces 'could maybe be proven with SLEC' by an open-source proof under a stated invariant.

3 Conditions

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

cv32e40p_controller.sv:640 condition hwlp_end1_eq_pc && !hwlp_counter1_gt_1 (with is_hwlp_body): reachable only if HWLoop1's end register lies inside an active HWLoop0 body (end1 in [start0+4, end0]), which the PULP HWLoop nesting rule (HWLoop0 inner, outer end >= inner end + 8) forbids. Under the invariant 'whenever hwlp_counter[0] > 1, hwlp_end_addr[1] is not in [hwlp_start_addr[0], hwlp_end_addr[0]+4]' plus well-formed loops (start < end) and no 32-bit wrap, the condition is formally unreachable (k-induction k=12, runs/v1-cv32e40p-1005/formal/p_line640_inv2_kind; z3 runs/v1-cv32e40p-1005/smt). Hardware does not enforce the rule (bounded cover reaches it in 7 cycles, runs/v1-cv32e40p-1005/formal/cover). Waive as software-illegal with that register-content invariant (it also covers stale HWLoop1 registers). Line 630 is the legal mirror and stays covered.

Checker used

checker_1005.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/cover/sby.log	present	cover PASS (128.2 s): line 630 and 640 taken arms at step 7; the hole condition (end1==pc+4, counter1<=1, is_hwlp_body) at step 7 -> reachable by hardware with illegal/stale registers
E2	counterexample	formal/cover/job/engine_0/trace0.vcd	not kept	hole witness: pc=0xffffffff inside loop0 [0xffffffff, 0x0 (wrapped)] cnt0=0xffff, end1=0x0=pc+4, cnt1=0 (cover_hole_line640_hw_only.png)
E3	smt-model	smt/with_invariant.out	present	z3 unsat: line-640 condition + is_hwlp_body impossible under end1 >= end0+8, start0 >= start1, no wrap
E4	smt-model	smt/with_disjoint.out	present	z3 unsat also for disjoint sequential loops (loop1 after loop0, non-empty bodies)
E5	smt-model	smt/without_invariant.out	present	z3 sat without the rule: end1 = pc+4 inside an active loop0, cnt1 = 0
E6	proof	formal/p_line640_inv2_kind/sby.log	present	k-induction (smtbmc yices, k=12) PASS (966 s) under: end1 not in [start0, end0+4] while counter0>1; start<end for active loops; no address wrap
E7	counterexample	formal/p_line640_inv_kind/job/engine_0/trace.vcd	not kept	v2 attempt (register rule only) FAIL: wrap witness start0=0xffffffff > end0=0xffffffff8; led to the well-formedness assumptions
E8	proof	formal/p_line640/sby.log	present	negative result kept: unconstrained abc pdr TIMEOUT 30 min (property false without invariant)
E9	proof	formal/p_line640_inv/sby.log	present	negative result kept: abc pdr with invariant did not converge in 2.4 h; k-induction used

Paths are relative to runs/v1-cv32e40p-1005. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.
- smt-model. A satisfying assignment from the SMT solver for the stated query.

Artefact(s) E2, E7 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

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Limitations and scope

- The waiver holds only while software obeys the rule stated in Action. If software can violate that rule, the waiver does not hold and the code is reachable.
- This is an agent investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

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Provenance and measurement

Run directory	runs/v1-cv32e40p-1005
Title source	Title and configuration read back from reports/issues/cv32e40p_issue_1005.pdf, the previously published report for this issue.
Inputs shown	title, body(links stripped), labels
Human record	Dolphin Design 2024 (issue body): waived in v2 by test reasoning (HWLoop1 cannot be nested in HWLoop0); 'could maybe be proven using SLEC'
Agent confidence	0.75. This is the investigating agent's own estimate. It is not a measurement and not a reliability figure.
Iterations	3
Run by	Claude directly (no headless agent)

Measured tool and step time

Step	Measured
read issue + locate line 640 + is_hwlp_body analysis	6.0 min (360 s)
v1 assert + 3 covers	4.0 min (240 s)
z3 with/without/disjoint	60.0 s
cover: hole reached, read witness	2.1 min (128 s)
v2 invariant assert: pdr no convergence, k-ind wrap CEX	17.6 min (1055 s)
v3 well-formed + no-wrap, k-induction PASS	16.1 min (966 s)
report	7.0 min (420 s)
Total	53.8 min (3229 s)

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

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